Operating-Condition-Embedded three-Dimensional (OCE-3D) Shaft Orbital Dataset for Fault Diagnosis of Rotating Machinery

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1 OCE-3D Dataset Overview

The Operating-Condition-Embedded Three-Dimensional (OCE-3D) shaft-orbit dataset is an experimental benchmark for fault diagnosis of journal-bearing rotor systems under varying rotational speeds. It was collected from a fault-seeded Journal-Bearing Rotor System (JBRS) and includes raw shaft-displacement signals, rotational-speed records, generated orbit samples, hierarchical fault labels, and corresponding fault-parameter information. The dataset covers baseline, unbalance, misalignment, and combined unbalance–misalignment conditions. Each fault condition is physically implemented through discrete hardware configurations and annotated using a type–subclass–size hierarchy. The OCE-3D dataset was constructed to support condition-aware shaft-orbit diagnosis. By embedding rotational speed into the sample representation, OCE-3D samples describe orbit evolution over a specified speed range rather than a single-speed planar trajectory. The dataset can be used for fixed-speed classification, cross-speed diagnosis, hierarchical fault recognition and transfer learning.

1.1 Label Hierarchy and Naming Convention

Following ISO 21940 [1] and ISO 14691 [2], the operating states of the JBRS are grouped into four high-level conditions with corresponding fault subclasses:

- Baseline Condition (BC), which represents the balanced reference state;
- Unbalance (UB), with Static Unbalance (SU), Moment Unbalance (MU), and Compound Unbalance (CU) as three subclasses;
- Misalignment (MA), with Angular Misalignment (AM), Parallel Misalignment (PM), and Compound Misalignment (CM) as three subclasses;
- Unbalance–Misalignment Condition (UMC), where unbalance and misalignment are imposed simultaneously.

Table 1 summarizes the correspondence between experimental conditions and hierarchical fault labels. As shown in the table, the dataset adopts a three-level label structure. Level-1 denotes the high-level condition or fault type, Level-2 denotes the corresponding physical fault subclass, and Level-3 denotes the discrete group or fault-size realization. For conditions without an additional subclass, the Level-2 field is marked as "–". The baseline condition is denoted by BC-1 to BC-4, representing four balanced reference groups rather than fault-size levels. For fault conditions, the four discrete size groups are denoted as S1–S4 and correspond to reproducible hardware configurations rather than continuously varying degradation stages.

For fault subclasses, the complete label follows the format `<Type>-<Subclass>-<Size>`. For example, `MA-AM-S3` denotes a misalignment fault (MA), the angular-misalignment subclass (AM), and the third fault-size group (S3). For the combined unbalance–misalignment condition, no separate subclass is defined, and the labels are written as UMC-S1 to UMC-S4.

Table 1: Experimental conditions and hierarchical fault labels.

Condition	Sub-condition	Size Group	Fault label: Type (Level-1)	Fault label: Subclass (Level-2)	Fault label: Severity (Level-3)
Baseline Condition (BC)	–	4	BC	–	BC-1 BC-2 BC-3 BC-4
Unbalance (UB)	Static Unbalance (SU)	4	UB	UB-SU	UB-SU-S1 UB-SU-S2 UB-SU-S3 UB-SU-S4
Unbalance (UB)	Moment Unbalance (MU)	4	UB	UB-MU	UB-MU-S1 UB-MU-S2 UB-MU-S3 UB-MU-S4
Unbalance (UB)	Compound Unbalance (CU)	4	UB	UB-CU	UB-CU-S1 UB-CU-S2 UB-CU-S3 UB-CU-S4
Misalignment (MA)	Angular Misalignment (AM)	4	MA	MA-AM	MA-AM-S1 MA-AM-S2 MA-AM-S3 MA-AM-S4
Misalignment (MA)	Parallel Misalignment (PM)	4	MA	MA-PM	MA-PM-S1 MA-PM-S2 MA-PM-S3 MA-PM-S4
Misalignment (MA)	Compound Misalignment (CM)	4	MA	MA-CM	MA-CM-S1 MA-CM-S2 MA-CM-S3 MA-CM-S4
Unbalance–Misalignment Condition (UMC)	–	4	UMC	–	UMC-S1 UMC-S2 UMC-S3 UMC-S4

Table 2 summarizes the sample distribution of the generated OCE-3D dataset. The samples are generated in a group-balanced manner, with 96 training samples, 20 validation samples, and 24 test samples assigned to each fault group. Because BC and UMC each contain four groups, while MA and UB each contain twelve groups, the sample numbers of MA and UB are three times those of BC and UMC. Overall, the dataset contains 3072 training samples, 640 validation samples, and 768 test samples.

1.2 Fault Parameters Corresponding to Each Label

Table 3 gives the complete correspondence between hierarchical labels and hardware-realizable fault parameters in the OCE-3D orbit dataset. The Level-1, Level-2, and Level-3 labels denote the fault type, fault subclass, and discrete group or fault-size realization, respectively. For the baseline condition, BC-1 to BC-4 represent balanced reference groups rather than fault-size levels.

The listed parameters describe how each fault label is physically implemented on the JBRS. The axial center-of-mass offset and radial center-of-mass offset characterize the axial position and radial eccentricity of the equivalent mass distribution, respectively. The mass change percentage describes the relative change in the tunable disk mass configuration. The axial parallel offset and angular misalignment describe the imposed parallel and angular alignment deviations. A value of 0 indicates that the corresponding parameter is not intentionally introduced for that label, whereas “Negligible” indicates that the parameter is not a target fault variable and remains sufficiently small under the corresponding hardware configuration.

Because the JBRS faults are implemented using discrete adjustment components, the listed parameters

Table 2: Sample distribution of the OCE-3D dataset.

	Each group	BC	MA	UB	UMC	Total
Train	96	384	1152	1152	384	3072
Val	20	80	240	240	80	640
Test	24	96	288	288	96	768

should be interpreted as reproducible hardware settings rather than continuous design variables. These settings are determined by available shims with thicknesses of 0.05, 0.10, 0.25, 0.50, and 0.60 mm, a 2011.32 g tunable disk, and 3.62 g screws with 0.10 g nuts installed at 18 circumferential holes. Accordingly, the Level-3 labels denote discrete experimental realizations rather than continuously increasing degradation stages. This design ensures that each dataset label is physically instantiated rather than nominally assigned.

1.3 Sample Generation Protocol

The OCE-3D samples are generated by embedding rotational speed into the shaft-orbit representation. As defined in Table 4, the rotational speed ω is assigned to the operating-condition axis, while D_x and D_y denote the horizontal and vertical shaft-displacement components, respectively. In the rendered OCE-3D space, ω is mapped to the X-axis, D_y to the Y-axis, and D_x to the Z-axis. The speed coordinate is rescaled to a displacement-comparable range, whereas fixed displacement limits are used for D_x and D_y to avoid orbit-shape differences caused by inconsistent axis ranges. Table 5 summarizes the fixed image-rendering parameters. Using identical resolution, viewing angle, color, line width, and file format across all sample types ensures that the generated images mainly differ in fault-induced orbit geometry and speed-dependent orbit evolution, rather than in rendering style.

Table 6 lists the deterministic generation settings for the training–validation pool. The five sample types cover different speed-window lengths, sliding steps, and downsampling intervals, allowing the generated samples to represent orbit evolution at multiple operating-condition resolutions. The sample quantities in the table are defined per fault group and sum to the training and validation samples assigned to each group.

Table 7 gives the test-generation ranges. In contrast to the fixed training–validation settings, the test samples are generated from unseen combinations of downsampling interval, speed-window length, and speed range. This design evaluates whether a diagnostic model remains robust to changes in sample-generation settings within the prescribed operating-speed range.

After the coordinate mapping, sample-generation settings, and image-rendering parameters are defined, representative OCE-3D samples are shown in Fig. 1. The figure illustrates how different fault labels and data splits are reflected in the generated orbit samples. The columns correspond to different Level-1 fault categories, while the rows compare samples from the training and test subsets. The examples also show that training and test samples follow the same OCE-3D representation principle but may differ in their starting speed, speed-domain length, and downsampling factor according to the generation protocol defined above.

Each generated image filename contains three logical parts: the fault label, the dataset split, and the sample-generation parameters. It follows the format

```
<fault label>_<split>_<start speed>_<sample length>_<downsampling factor>.png
```

where the start speed and sample length are expressed in rpm, and the downsampling factor is dimensionless.

For example,

```
MA-AM-S3_train_1400_200_16.png
```

denotes a training sample under the misalignment category, angular-misalignment subclass, and fault-size level S3. The sample is generated from a speed segment starting at 1400 rpm, with a speed-domain length of 200 rpm and a downsampling factor of 16.

Table 3: Detailed experimental fault conditions, physical parameters, and hierarchical label configuration.

Fault label			**Experimental parameters**				
Type (Level-1)	Subclass (Level-2)	Severity (Level-3)	Axial Center Mass Offset (mm)	Radial Center of Mass Offset (mm)	Mass Change Percentage (%)	Axial Parallel Offset (mm)	Angular Misalignment (°)
BC	–	BC-1	0	0	0	0	0
		BC-2	0	0	0	0	0
		BC-3	0	0	0	0	0
		BC-4	0	0	0	0	0
UB	UB-SU	UB-SU-S1	0	0.0702	0.290	0	0
		UB-SU-S2	0	0.0699	0.123	0	0
		UB-SU-S3	0	0.1209	0.246	0	0
		UB-SU-S4	0	0.0702	0.145	0	0
	UB-MU	UB-MU-S1	0	0	0.339	0	Negligible
		UB-MU-S2	0	0	0.236	0	Negligible
		UB-MU-S3	0	0	-0.012	0	Negligible
		UB-MU-S4	0	0	-0.420	0	Negligible
	UB-CU	UB-CU-S1	63.41	0.0536	0.114	0	Negligible
		UB-CU-S2	63.41	0	0.307	0	Negligible
		UB-CU-S3	63.41	0.0719	0	0	Negligible
		UB-CU-S4	63.41	0.0702	-0.145	0	Negligible
MA	MA-AM	MA-AM-S1	0	0	0	0	0.00716
		MA-AM-S2	0	0	0	0	0.00517
		MA-AM-S3	0	0	0	0	0.00119
		MA-AM-S4	0	0	0	0	0.01034
	MA-PM	MA-PM-S1	0	0	0	0.050	Negligible
		MA-PM-S2	0	0	0	0.100	Negligible
		MA-PM-S3	0	0	0	0.087	Negligible
		MA-PM-S4	0	0	0	0.103	Negligible
	MA-CM	MA-CM-S1	0	0	0	0.087	0.00716
		MA-CM-S2	0	0	0	0.100	0.00716
		MA-CM-S3	0	0	0	0.050	0.00716
		MA-CM-S4	0	0	0	0.087	0.01592
UMC	–	UMC-S1	0	0.0702	-0.15	0.050	0.00716
		UMC-S2	0	0.0702	-0.15	0.100	0.00716
		UMC-S3	0	0.0536	-0.11	0.043	0.00716
		UMC-S4	63.41	0.0702	-0.15	0	0.02003

Table 4: Coordinate parameters for OCE-3D orbit generation.

Parameter	X-axis	Y-axis	Z-axis
Variable	ω	D_y	D_x
Unit	rpm	mm	mm
Axis scale	500	1	1
Axis range	Adaptive	[1.8, 2.2]	[1.8, 2.2]

Table 5: Image parameters.

Parameter	Value
Viewing angle	(−37.5, 30)
Image size	832 × 832
Orbit color	Black
Line width	0.05
Format	PNG

Table 6: Training- and validation-set sample-generation parameters.

Sample type	Downsampling interval	Sample length (rpm)	Sliding step (rpm)	Sampling interval (rpm)	Sample quantity
Sample type 1	128	1000	100	[1000, 3000]	11
Sample type 2	50	500	100	[1000, 3000]	16
Sample type 3	32	400	100	[1000, 3000]	17
Sample type 4	20	300	50	[1000, 3000]	35
Sample type 5	16	200	50	[1000, 3000]	37

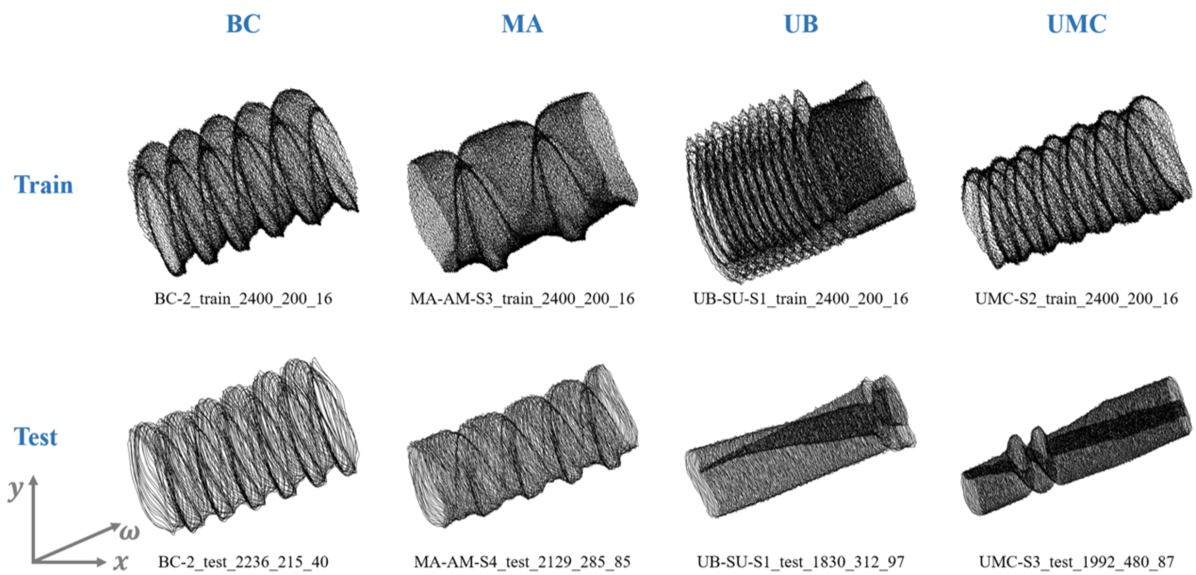


Figure 1: OCE-3D shaft-orbit samples.

Table 7: Test-set sample-generation parameters.

Parameter	Value
Downsampling range	[21, 49], [51, 99]
Sample-length range (rpm)	[201, 299], [301, 399], [401, 499], [501, 599]
Sampling-interval range (rpm)	[1000, 2000], [1500, 2500], [2000, 3000]

2 Fault-Seeding Experiments

2.1 Fault-Seeded JBRS Test Rig

The OCE-3D dataset was collected on a laboratory-scale fault-seeded JBRS test rig. As shown in Fig. 2, the test rig consists of a drive–control module, a fault-injectable bearing–rotor module, a tunable counterweight disk, journal bearings, eddy-current displacement sensors, a tachometer, and a data acquisition interface. The system supports controlled speed variation, radial shaft-displacement measurement, and repeatable physical implementation of unbalance, misalignment, and combined faults.

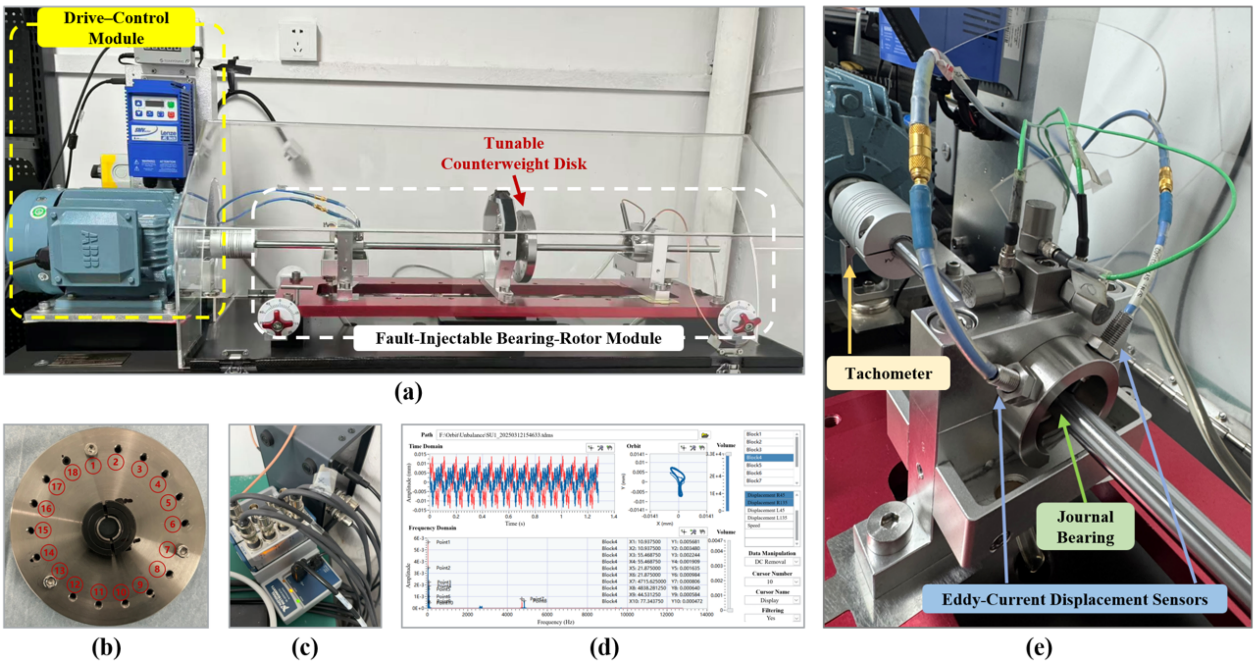


Figure 2: Fault-seeded journal-bearing rotor test rig.

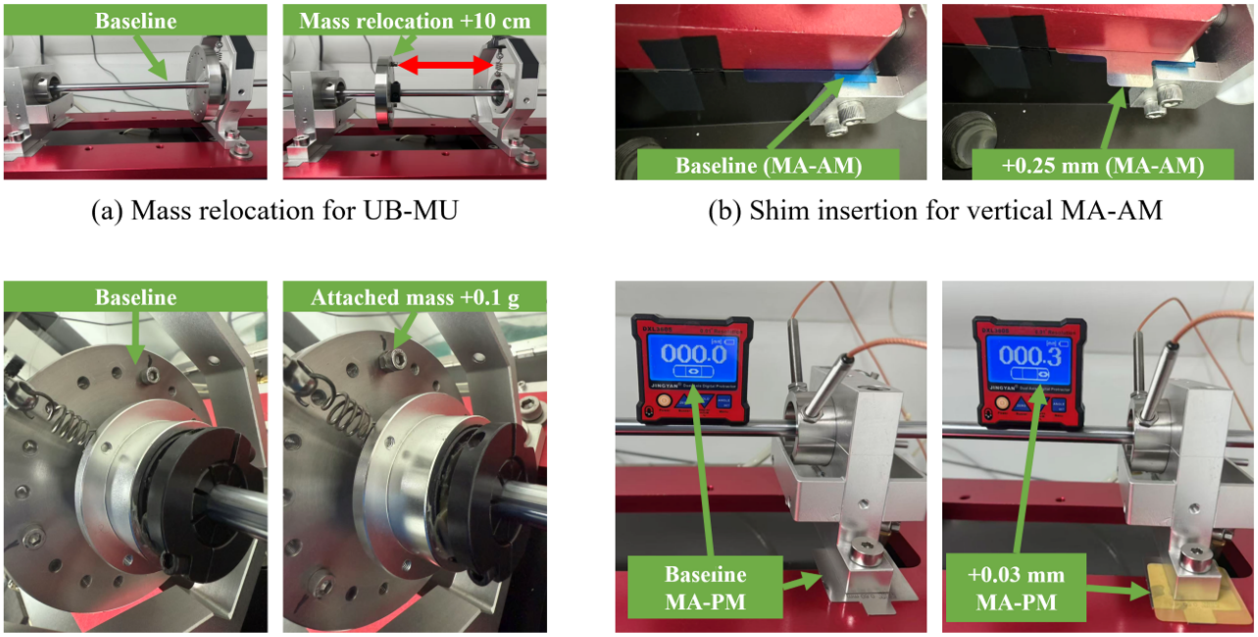
The drive–control module regulates the rotor speed during data acquisition. The tunable counterweight disk provides a configurable platform for unbalance seeding, while the bearing supports and alignment components allow misalignment to be introduced through mechanical adjustment. Two orthogonal eddy-current displacement sensors measure the horizontal and vertical shaft motions, from which shaft orbits are constructed. The tachometer provides rotational-speed information, which is further used as the operating-condition coordinate in OCE-3D sample generation.

2.2 Fault-Seeding Design and Discrete Parameter Settings

The fault labels in the dataset are determined by physical fault-seeding operations rather than by post hoc class assignment. Unbalance faults are implemented by changing the mass distribution of the tunable disk, whereas misalignment faults are implemented by inserting shims to adjust the relative position or angle of the shafting system. The combined unbalance–misalignment condition is generated by applying both mechanisms simultaneously.

Figure 3 illustrates representative hardware operations used to instantiate the fault labels. For moment unbalance, the mass position is relocated along the axial direction of the rotor. For static unbalance, additional masses are attached to specified circumferential positions on the tunable disk. For angular and parallel misalignment, shims with prescribed thicknesses are inserted at the bearing-support locations to introduce controlled alignment deviations.

Because these operations are constrained by available mechanical components, the resulting fault parameters are discrete rather than continuously adjustable. The shim thicknesses, screw and nut masses, disk



(a) Mass relocation for UB-MU

(b) Shim insertion for vertical MA-AM

(c) Additional mass attachment for UB-SU

(d) Shim insertion for vertical MA-PM

Figure 3: Representative fault-seeding operations.

geometry, and circumferential mounting holes jointly determine the realizable fault configurations. Therefore, the size labels S1–S4 should be interpreted as reproducible hardware realizations, not as strictly monotonic degradation stages.

3 Usage Notes

3.1 Intended Applications

The dataset is intended for shaft-orbit-based fault diagnosis of rotating machinery, especially under operating-speed variation. It can be used for fixed-speed fault classification, cross-speed diagnosis, hierarchical fault recognition, representation comparison among 2D, T-3D, and OCE-3D samples, and transfer-learning studies across operating conditions.

The dataset is not intended to represent a continuous degradation process. The size labels are discrete hardware realizations, so they should not be directly interpreted as remaining useful life stages or monotonic damage progression. Because the data are collected from a controlled laboratory JBRS, direct transfer to industrial machinery should be performed with appropriate validation or recalibration.

3.2 Recommended Model Types and Benchmark Settings

For the generated orbit images, vision models are recommended as the primary modeling framework. In particular, classification backbones from modern vision networks are suitable choices, because the generated OCE-3D samples are image-based representations with explicit fault-label annotations. The benchmark is not restricted to a specific architecture; CNN-based, residual-network-based, Transformer-based, or other image-recognition backbones may be used according to the diagnostic task and model-comparison objective.

For benchmark evaluation, the provided train, validation, and test splits should be kept unchanged whenever possible. The training and validation subsets should contain the same sample types defined in Table 6. If multiple sample types are pooled and randomly split without sample-type stratification, the resulting distribution mismatch may affect training stability and validation reliability. Therefore, any random split should be stratified by both fault label and sample type. The test subset should be reserved for final evaluation, with performance reported at the fault-type, subclass, and size-group levels.

3.3 Citation, License, and Contact

Users of the dataset should cite both the dataset record and the associated paper, "From 2D to 3D: A Novel Shaft Orbit Design Methodology and Its Application to Cross-Condition Fault Diagnosis of Bearing–Rotor System". The recommended citation format will be provided together with the paper when it is published. If a dataset DOI is assigned, it should be used as the primary dataset identifier.

Detailed documentation, raw data, sample-generation code, contact information, and future updates will be provided through the following GitHub repository:

`https://github.com/ZhaorongL/OCE-3D-Shaft-Orbit-Dataset`

 Questions about the dataset, label definitions, sample-generation protocol, or code usage can be directed to the dataset maintainers:

`ruandiwang@zju.edu.cn`; `lizhaorong@zju.edu.cn`.

References

[1] ISO 21940: Mechanical vibration – Rotor balancing.

[2] ISO 14691: Petroleum, petrochemical and natural gas industries – Flexible couplings for mechanical power transmission.